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RESEARCH-ARTICLE

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Hardware-aware Calibration Protocol for Quantum Computers

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Abstract

Calibration of a quantum computer is the process of optimizing its control parameters to ensure the accurate implementation of quantum gates. It remains a critical challenge in scaling quantum computers. Existing calibration methods take a generalized approach that focuses on the trade-off between calibration time and fidelity. However, these methods lack the awareness of hardware differences among physical qubits and an elaborate design of parallel calibration. In this paper, we introduce a fine-grained calibration protocol that contains three calibration policies for hardware differences and a method to enable parallel calibration. We begin by profiling qubit pairs to evaluate their responses to different waveform candidates. Based on profiling results, we determine the best calibration policy for the quantum computer, which is the first part of the calibration protocol. The second part of our protocol is to use graph traverse to enable parallel calibration by identifying compatible calibration operations. We validate our protocol through intensive experiments on real quantum machines with up to 127 qubits. Our experimental results demonstrate a 1.84× reduction in terms of the medium of the two-qubit gate error rate, 1.26× reduction in pulse duration, an 8× to 25× reduction in total calibration overhead compared with sequential calibration, an average of 2.12× further reduction in total calibration overhead owing to profiling policy, double of the quantum volume, and a 2.0× to 2.3× reduction in error per layered gate. The proposed protocol emphasizes the importance of hardware-aware and parallel calibration and advances current quantum computers towards fault-tolerant quantum computing.

*Both authors contributed equally to this research.

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CCS Concepts

• **Computer systems organization** → **Quantum computing**.

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1 Introduction

Recent advances in quantum computing have demonstrated impressive scaling in qubit numbers, with leading platforms now reaching hundreds of physical qubits [1, 10]. However, achieving high-fidelity quantum operations on these large-scale systems remains a critical challenge [4, 11, 49, 57]. While single-qubit gate fidelities often exceed 99.9% [6, 15], two-qubit gate fidelities on multi-qubit, fixed-frequency superconducting chips typically lie around 99% [14, 52], significantly higher than the 0.1% error rates on specialized two-qubit devices. As one of the leading quantum computing platforms, superconducting quantum systems face challenges in maintaining high-fidelity operations as the system scales up, which signifies the importance of a good calibration protocol.

Superconducting quantum computers operate at extremely low temperatures in high-vacuum environments to maintain quantum coherence [24]. These systems rely on microwave pulse control for qubit manipulation and measurements. The microwave signals define the system Hamiltonian and the qubit evolution, while the reflected signals of measurement pulses enable quantum state measurements. The quality of quantum operations depends on the precise calibration of these control systems [7, 19]. Precise control involves two key aspects: generating accurate microwave pulses and performing regular calibrations to compensate for system drift and external perturbations. The control parameters include pulse amplitudes, frequencies, and durations, which must

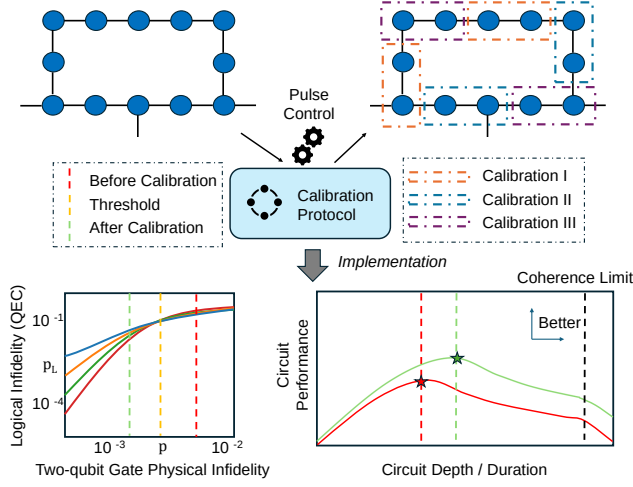


Figure 1: Overview of the proposed hardware-aware calibration protocol and its impact.

be carefully calibrated based on measurements of current qubit properties [13, 22, 51]. To maintain optimal operation conditions and keep error rates within acceptable ranges, quantum hardware providers like IBM perform calibration routines at both hourly and daily intervals [40]. Regular calibration ensures that qubits perform the designed operations. While individual gate errors may appear insignificant in isolation, their cumulative effect becomes significant as circuit depth increases and the number of gates grows. As these errors accumulate, they can significantly degrade the overall fidelity of the quantum circuit, and eventually make the execution of quantum algorithms unreliable [16, 23, 41, 54, 55].

Calibration is also critical for Quantum Error Correction (QEC) since when effective calibration is applied to reduce the physical error rates below a critical threshold, QEC can exponentially suppress logical errors by increasing the code distance [8, 44]. Recent advancements, such as below-threshold operation of surface codes [37] and hardware-efficient schemes like bosonic cat qubits [56], have shed light on the potential of QEC in the near future. These advances have driven extensive experimental efforts to optimize QEC protocols in real quantum systems. Related works [5, 25, 35, 42] demonstrate that achieving high-fidelity operations for practical QEC demands precise calibration. However, traditional calibration methods face significant challenges. First, they often ignore differences in qubits' properties. Second, the extended duration of calibration—spanning hours or days [40]—introduces system drift that affects the system performance. Finally, existing approaches emphasize a trade-off between calibration time and fidelity but neglect hardware-specific optimizations and the strict timing requirements of QEC protocols. These limitations are particularly severe in calibrating cross-resonance (CR) gates which are essential for two-qubit operations in quantum processors. As QEC becomes a cornerstone of fault-tolerant quantum computing, it imposes rigorous demands on gate fidelities and control accuracies.

A key challenge in CR gates arises from unwanted excitations of the control qubit due to off-resonant drives. The standard derivative removal by adiabatic gate (DRAG) technique addresses this issue

for single transitions [12, 33], but quantum systems often exhibit multiple transition pathways that need simultaneous control. Multi-derivative DRAG offers a solution by applying recursive corrections to suppress multiple transitions simultaneously [27]. This method adds derivative terms to the control pulse that target specific energy transitions between quantum states. While this approach can enhance gate fidelity, its effectiveness depends on system parameters such as the frequency difference between qubits. These implementation considerations motivate the need for a systematic approach to optimize calibration across multiple qubit pairs simultaneously.

To optimize the calibration process across qubit pairs, we introduce a novel calibration protocol with three calibration policies: (1) **Brute-force Clustering**, where qubit pairs are grouped based on key physical properties, representative pairs are selected for calibration, and the optimized Echoed Cross-Resonance (ECR) waveforms are generalized; (2) **Topology-oriented Representative**, which leverages the regular patterns in heavy-hex lattice topologies by classifying qubit pairs according to positions within unit cells, calibrates representatives for each position, and generalizes the optimized waveforms; and (3) **Hardware-oriented Policy**, which incorporates system knowledge and hardware limitations, such as frequency detuning and decoherence times, to select the most efficient and practical waveform strategies for each qubit pair.

Our policies also offers a practical balance between calibration accuracy and computational efficiency, which is particularly important for scaling up quantum processors. However, successful local optimization of qubit pairs is only the first step toward processor-wide calibration. We address processor-wide calibration by treating the quantum processor as an undirected graph G , where nodes represent qubits and edges represent coupled qubit pairs. To minimize calibration time, we develop a parallel calibration protocol that partitions the coupling graph into calibration subgraphs. Each subgraph consists of edges that can be calibrated simultaneously without mutual interference, achieved by maintaining a minimum distance of two between concurrent calibrations. This parallelization strategy significantly accelerates the calibration process while preserving calibration accuracy. For instance, in a 127-qubit heavy-hex architecture, our approach enables simultaneous calibration of up to 38 qubit pairs distributed across five subgraphs, substantially reducing the total calibration time compared to sequential methods.

We illustrate the brief overview of our protocol in Figure 1. Our protocol optimizes calibration policies and assigns different pulse envelopes to qubit pairs as the colored blocks in the top-right of Figure 1. The bottom-left of Figure 1 shows the effectiveness of quantum error correction at various code distances, represented by different colored curves. Our protocol achieves two key improvements: it reduces two-qubit gate physical error rates below the threshold, enabling effective QEC and significantly improving logical error rates; and it optimizes system performance by reducing calibration overhead, shortening circuit duration, and enhancing overall circuit fidelity compared to conventional calibration approaches, as shown in the bottom-right of Figure 1.

In this work, we introduce a novel fine-grained calibration protocol with three major contributions:

- A novel calibration protocol with multiple components: 1) Three calibration policies to assign waveform candidate to

qubit pairs according to their profiling results. 2) A graph-based parallelization algorithm for scaling up the protocol that performs the calibration simultaneously.

- The first large-scale implementation of multi-derivative DRAG and direct CR operations on real quantum machines, which advances quantum control techniques in practice.
- Extensive experimental results with up to 127 qubits show a $1.84\times$ reduction in terms of the medium of the two-qubit gate error rate, $1.26\times$ reduction in pulse duration, up to an $8\times$ to $25\times$ reduction in total calibration overhead compared with sequential calibration, an average of $2.12\times$ further reduction in total calibration overhead owing to profiling policy, double of the quantum volume, and a $2.0\times$ to $2.3\times$ reduction in error per layered gate.

2 Background

2.1 Calibration and Characterization in Quantum Gates

In quantum computing, qubits and quantum gates exhibit extreme sensitivity to external disturbances [39]. Calibration and characterization thus serve as essential procedures in quantum hardware experiments. For superconducting qubits, quantum gates operate through precise control pulses, which require careful tuning of multiple parameters: drive frequency, amplitude, phase, and duration [28]. Each parameter affects the qubit's dynamics directly. For example, small amplitude deviations can cause over-rotations or under-rotations of qubits. These errors accumulate and degrade overall circuit performance and fidelity as operations scale. Therefore, amplitude adjustments must be performed through iterative, fine-grained calibration routines. This process requires amplitude tuning in small increments to achieve the desired qubit state transition, which aligns actual gate behavior more closely with the intended ideal operation. Moreover, these parameters need continuous optimization to maintain stability over time, particularly in the presence of inherent hardware variations such as frequency drifts and crosstalk. To optimize the calibration of qubits, we need to understand the properties of quantum gates on the qubits. Characterization techniques reveal crucial insights into the performance of quantum gates in physical devices [38]. Quantum Process Tomography (QPT) is an approach that constructs the entire quantum process matrix for a certain quantum gate [32]. This method requires qubit preparation in various input states, gate application, and measurement of output states to characterize the gate's behavior fully. However, QPT has bad scalability and it's not applicable to larger systems. Randomized Benchmarking (RB) [21] presents a more efficient alternative for characterization. This statistical approach determines average gate error rates through random gate sequences applied to qubits. Modern implementations often incorporate interleaved randomized benchmarking [31], where a specific gate of interest alternates with random Clifford operations, enabling characterization of individual gate performance within the broader context of circuit operation. Three metrics are utilized to evaluate the calibration process, including gate fidelity, gate duration, and calibration cost, which is measured as the total quantum seconds of calibration.

2.2 Echoed CR

The Echoed Cross Resonance (Echoed CR) pulse [2] is an existing implementation method for two-qubit gates in superconducting quantum computing, particularly for the CNOT gate implementation. The standard CR pulse has multiple Hamiltonian interactions: ZX , ZZ , IZ , IX , IY , and ZY . The ultimate goal of calibration is to tune up a suitable ZX interaction to generate a precise 90 degree rotation while minimizing other unwanted interactions[43]. The Echoed CR pulse is built upon the CR pulse by adding an "echo" mechanism to reduce phase errors and unwanted couplings during gate operation. This echoing procedure eliminates unwanted Hamiltonian terms and preserves the desired ZX interaction. To calibrate the Echoed CR pulse, we need to calibrate the CR pulse within it. The calibration process first focuses on the selective cancellation of remaining ZY , IX , IY terms through iterative adjustments of CR and target qubit drive pulses. The system performs two tomography experiments per calibration round to refine the pulse parameters. Hamiltonian tomography [43] is adopted to measure the target qubit's response with various time intervals. The data from Hamiltonian tomography provides coefficient $v(t)$ and enables the decomposition of the Hamiltonian into Pauli terms[30]:

$$\hat{H}(\Omega_{\text{CR}}, \Omega_{\text{T}}) = v_{ZX}\hat{Z}\hat{X} + v_{ZY}\hat{Z}\hat{Y} + v_{IX}\hat{I}\hat{X} + v_{IY}\hat{I}\hat{Y} + v_{ZI}\hat{Z}\hat{I}. \quad (1)$$

The suppression of ZZ interactions (also known as ZZ coupling strength) requires precise IY DRAG calibration, which modifies the control pulse with an additional Gaussian derivative component and lifting[27]. This process involves sampling three different IY DRAG amplitudes to identify zero points of the ZZ coupling strength through linear fitting. The final implementation uses an echoed pulse sequence. This sequence consists of a primary CR pulse followed by a reversed CR pulse. For an echoed CNOT gate, each CR pulse must last one-eighth of the period, as the target qubit rotates in opposite directions based on the control qubit state.

2.3 Direct CR

Direct CR [9] is more expensive in terms of calibration cost against Echoed CR because of it based on Echoed CR with additional technologies involved. The calibration process begins with adjustment on target qubit signals to satisfy the condition $v_{IX} = -v_{ZX}$ in Eq. 1. Through dividing the target oscillations corresponding to the control state, the coefficient v can be expressed as $v_{IX} = (\Omega_x^0 + \Omega_x^1)/2$ and $v_{ZX} = (\Omega_x^0 - \Omega_x^1)/2$, where $\Omega_{x,y}^{0,1}$ is the drive strength in along x or y under the control of 0 or 1. The prior condition ensures $\Omega_x^0 = 0$, indicating the target qubit rotates only when the control qubit exists in the $|1\rangle$ state. The adjustment is implemented through the iterative calibration described in the echoed CR calibration. The next phase addresses the problem of phase shift on control qubit, which can be seen as quantum state change due to applied electrical field, also know as the the Stark effect with the CR pulse[43]. This calibration step is the key difference from echoed CR calibration. In summary, the Echoed CR pulse automatically cancels phase shifts, but direct CR requires specific phase calibration. The phase calibration process employs two specific circuit configurations, as shown in Figure 2a and 2b. The circuit in Figure 2a implements a sequence of $2N$ uncalibrated CR pulses, each paired with an $R_Z(\phi)$ rotation

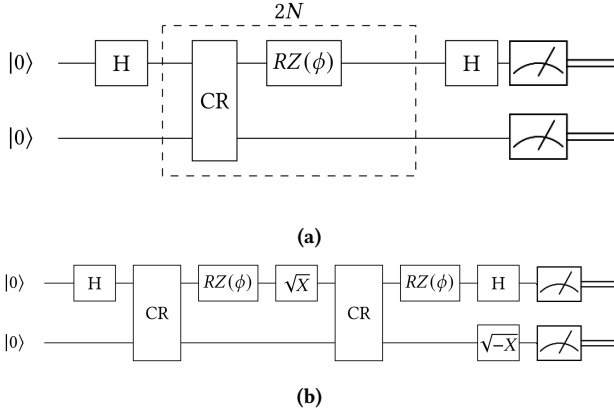


Figure 2: (a) Calibration circuit for Z phase for the direct CR. (b) Verification circuit for the direct CR.

on the control qubit, which resembles the Floquet calibration process [17]. The $2N$ sequence is inserted between 2 Hadamard gates in order to amplify the phase shift for measurement and calibration. The circuit returns to its initial state only when the CR pulses combined with the rotation produce a standard CNOT with a 0-degree or 180-degree phase shift. Figure 2b presents a verification circuit that confirms the implementation of a proper CNOT gate.

3 Motivation

Existing calibration methods face two major challenges: hardware inconsistencies and high calibration costs. While recent advancements propose solutions through noise-aware optimization, adaptive protocols, and real-time feedback, these approaches often introduce significant overhead in both computation and time resources. This overhead can reduce quantum processor availability for practical workloads. Therefore, an improved calibration protocol must achieve three key objectives: (1) address hardware-specific variations, (2) maintain high fidelity with reasonable latency, and (3) minimize calibration costs while ensuring system performance.

3.1 Hardware Variability in Superconducting Quantum Systems

Hardware limitations restrict the performance of superconducting quantum computers. These constraints result in weak qubit interaction strength and low quantum gate fidelity, which make precise calibration technology essential for reliable operation. The significant differences in relaxation time (T_1) and dephasing time (T_2) between qubit pairs create a critical challenge. Short decoherence times lead to rapid quantum information loss in qubits, which directly affects the fidelity of two-qubit operations. This inconsistency requires individual calibration protocols for each qubit pair and adds substantial complexity to system maintenance. Furthermore, fabrication variations and material differences cause wide frequency detuning values across qubit pairs. These inherent hardware variations mean that standardized calibration methods and uniform control pulses **cannot** achieve optimal performance across all qubit pairs, which results in reduced overall gate fidelity. This

hardware-specific behavior demands more sophisticated, tailored calibration approaches. We address this problem in Section 4.2.

3.2 Missing Performance Metrics in Traditional Calibration

Traditional calibration methods rely on single-gate fidelity as the primary performance metric, which provides an incomplete picture of quantum **system** performance. Advanced compilation techniques [29, 47, 48, 50] have revealed that optimizing individual gate fidelity alone fails to capture critical system-level effects, such as qubit idle times and decoherence during circuit execution. Quantum Volume addresses these limitations by providing a metric that includes both qubit quality and system performance. However, current calibration techniques continue to focus on optimizing isolated metrics like gate fidelity or latency, without incorporating metrics like Quantum Volume. This narrow focus on performance assessment leads to suboptimal calibration strategies that fail to capture the true capabilities of quantum devices. We demonstrate gate-level, calibration-level, device-level and application-level improvements of our protocol in the Section 5.

3.3 High-overhead and Infrequent Calibration

Current generalized calibration approaches face significant limitations due to the high costs associated with the calibration. These costs lead to infrequent calibration, which allows system drift and errors to accumulate over time. The problem is especially severe for two-qubit gates, which are essential for advanced quantum algorithms and error correction but require longer calibration times to maintain coherence. There is an inherent trade-off between more frequent calibration to improve gate fidelity and system downtime this calibration requires. On IBM quantum hardware, two-qubit gates need long calibration cycles to reach acceptable fidelity levels, results in extended downtimes. During these periods, system properties can drift, and this causes unexpected errors even in simpler single-qubit gates. Current calibration approaches do not perform calibration often enough to sufficiently mitigate these issues.

Experiments on IBM hardware demonstrate the consequences of insufficient calibration. Measurements showed error rates for single-qubit gates that reached as high as 10^{-2} when trying to excite qubits to the $|1\rangle$ state. These high error rates highlight the impact of infrequent calibration on gate fidelity. As circuits grow in size, errors from two-qubit gates and drifting system properties accumulate, and this significantly affects the computational results. After we monitor and profile the IBM's machine for weeks, we notice IBM's current calibration standards focus on weekly full calibration of only a limited number of qubit pairs. Daily measurements include phase calibrations for just a few pairs [40]. This approach allows substantial system drift to occur, as the minor fidelity improvements from more frequent calibration are considered unjustified given the high calibration costs. This significantly limits gate fidelity and computational accuracy, especially as quantum circuits become larger. We address the challenge of large-overhead calibration by parallel calibration as described in Section 4.3.

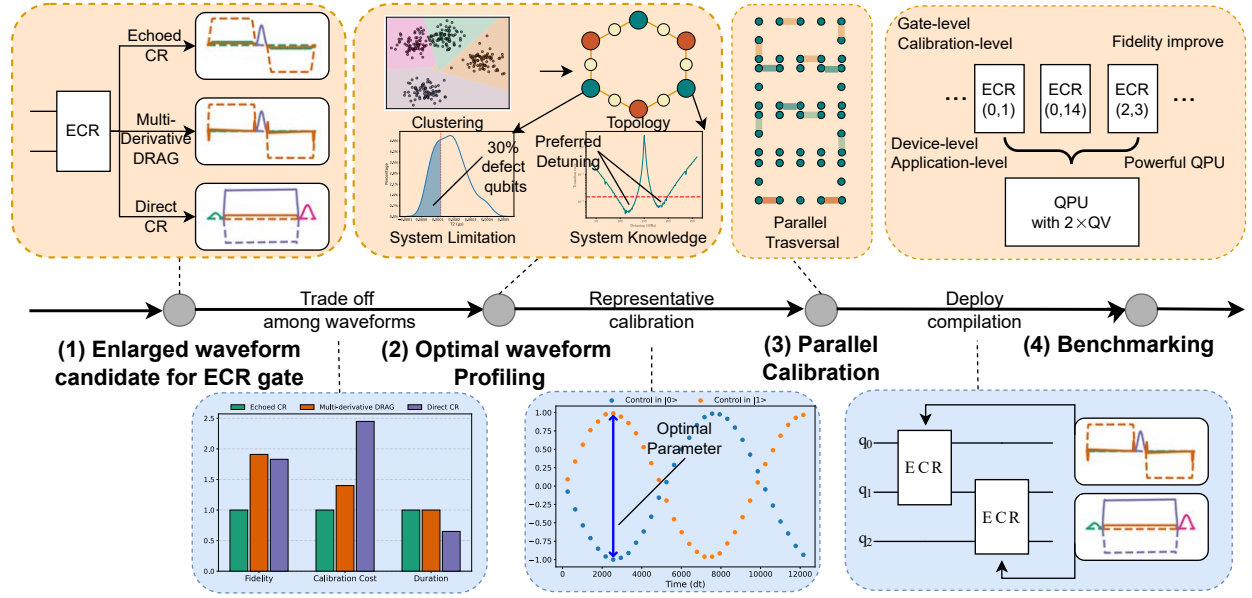


Figure 3: Overview of the proposed calibration protocol design. With enlarged pulse candidates for ECR gate, optimal waveform profiling is achieved by the proposed calibration policies. Parallel calibration is adopted to accelerate the calibration process. The effectiveness of the protocol is examined from gate-level, calibration-level, device-level, and application-level.

4 Protocol

In this section, we are introducing our proposed calibration protocol. The performance of current quantum hardware is leveraged and characterized through 4 steps, as depicted in Figure 3.

Enlargement of candidate waveform. Three distinct waveforms are developed, each capable of implementing the same basis two-qubit gate on current quantum hardware. These waveforms offer trade-offs among gate fidelity, calibration cost, and gate duration, providing flexibility in optimizing quantum operations based on system constraints.

Optimal pulse profiling. In this step, multiple policies are employed, aiming at profiling each qubit pair with its own optimal pulse waveform. The policies are designed upon the physical properties, topology similarity and system knowledge.

Parallel Calibration. After the optimal pulse is decided, a parallel graph traversal is implemented to calibrate all qubit pairs with limited time.

Benchmarking. The improvement brought about by optimal pulse profiling is characterized from gate-level, calibration-level, device-level, and application-level.

4.1 Multi-derivative Drag

Ideally, the CR interaction leads to a rotation in the target qubit status and leaves the control qubit unaltered. However, the off-resonant microwave drive may excite the control qubits in the operation, especially when the frequency detuning between the qubits is smaller than their anharmonicities. To overcome the unwanted excitations, a term proportional to the derivative of the

drive pulse is introduced, the DRAG pulse, given by $\Omega - i\frac{\dot{\Omega}}{\Delta}$, where Δ denotes the energy difference.

For the off-resonant CR drive, all the transitions among the three states ($|0\rangle$, $|1\rangle$ and $|2\rangle$) of the control transmon need to be avoided simultaneously. Therefore, multi-derivative DRAG is applied to suppress the undesired transitions by recursively applying the DRAG correction targeting three dominant transitions [27, 34]

$$\Omega_{\text{CR}}^{\text{P}} = \mathcal{F}_{\Delta_{21}}^{(1)} \circ \mathcal{F}_{\Delta_{10}}^{(1)} \circ \mathcal{F}_{\Delta_{20}}^{(2)}(\Omega) \quad (2)$$

where $\Omega = \mathcal{F}_{\Delta}^{(n)}(\tilde{\Omega}) := \left(\tilde{\Omega}^n - i\frac{d\tilde{\Omega}^n}{dt}\frac{1}{\Delta}\right)^{\frac{1}{n}}$ and Δ_{jk} denotes the energy difference between state $|j\rangle$ and $|k\rangle$. The circle operator applies the pulse shape transformation $\mathcal{F}$ on top of the last pulse output. The explicit formula for the DRAG pulse is given as: $\Omega_{\text{CR}}^{\text{P}} = \Omega_1 - i\frac{\dot{\Omega}_1}{\Delta_{10}}$, $\Omega_1 = \Omega_2 - i\frac{\dot{\Omega}_2}{\Delta_{21}}$, and $\Omega_2 = \sqrt{\Omega_3^2 - i\frac{2\dot{\Omega}_3\Omega_3}{\Delta_{20}}}$. The initial input pulse shape Ω_3 needs to be chosen so that the final pulse shape still starts and ends at zero [27].

Another major portion of the CR error derivatives from other dynamical operators in the two-qubit subspace that do not commute with the ideal dynamics ZX. For instance, the unwanted terms in Equation 1 can be measured by Hamiltonian tomography and calibrated away, as in the standard CR pulse [43]. The remaining terms, such as the ZZ and IZ error [30], are also removed by DRAG corrections on the target qubit and adjusting the detuning [27].

Despite the improvement in gate quality, the multi-derivative pulse introduces additional calibration overhead. The calibration of the DRAG correction requires more tomography experiments and hence longer calibration time. The waveform generated under the multi-derivative DRAG tends to be more complicated, therefore

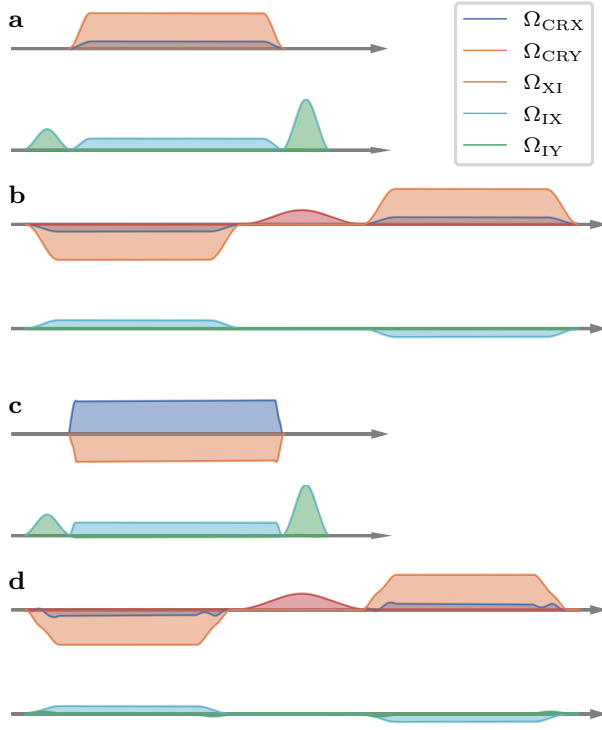


Figure 4: Examples of the CR pulse: the default pulse shape for direct (a) and echoed (b) CR as well as the multi-derivative direct CR (c) and echoed (d). The pulse amplitudes are rescaled for visualization purposes.

challenging the hardware configuration and resource optimization for current quantum hardware. During calibration experiments on real quantum hardware, a preprocessing error is sometimes detected due to the overwhelmingly complicated waveform. More importantly, through numerical simulations, previous work has proved that multi-derivative DRAG waveform achieves the most significant improvement in a certain range of qubit-qubit detuning, e.g. as shown in Figure 6. Consequently, applying multi-derivative DRAG may be unnecessary for qubits with frequency detunings outside of this interval.

4.2 Calibration Policy Design

The proposed approach involves three pulse waveforms, and a straightforward solution would be to conduct calibration for all three waveforms across every qubit pair. However, this method would be resource-intensive and time-consuming. Due to hardware limitations and external disturbances, modern quantum hardware experiences considerable error drift; specifically, qubit pairs typically reach an error level $5\times$ their initial value within approximately **20 hours**. To maximize fidelity while minimizing calibration time, an efficient policy should be developed to identify and calibrate the optimal waveform for each qubit pair.

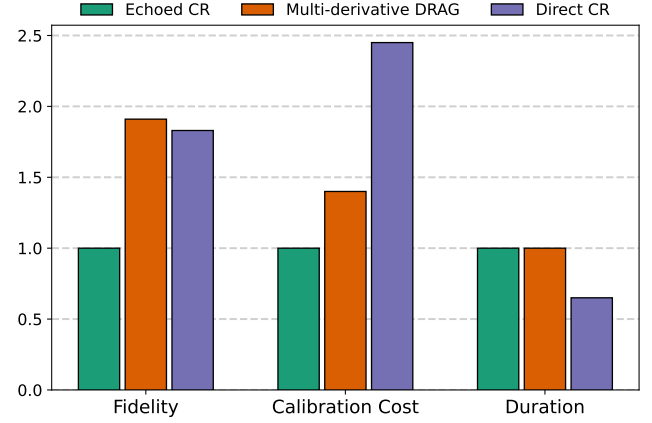


Figure 5: Fidelity, calibration cost, and duration trade-off among various ECR gate waveforms.

4.2.1 Waveform Candidates and Analysis. Based on previous discussions, any ECR gate could have multiple implementations, including the echoed CR waveform, the Multi-derivative DRAG waveform, and also the Direct CR waveform. Though the Direct CR waveform is calibrated towards the goal of realizing a CX operation, it could also be utilized to implement an ECR gates' waveform with an SX waveform and an X waveform on the target qubit. The cross-resonance part in the Direct CR can also be designed as containing multi-derivative parts. Therefore, the waveform candidates to realize an ECR gate are enlarged to four, as shown in Figure 4. However, calibrating the original direct CR waveform on real quantum hardware has been found to be exceedingly resource-intensive. As a result, in the following discussions, the Direct CR is implemented with multi-derivative parts.

Though various waveforms can be utilized to realize the same gate, a trade-off between gate fidelity, calibration cost, and duration exists among various waveforms. Multi-derivative DRAG and Direct-CR waveforms share a potential to achieve a higher fidelity than the echoed CR, while the prior introduces a 1.4 times increase in calibration cost, and the latter one requires a calibration cost as high as 2.8 times (Figure 5). However, the short duration of the Direct CR waveform is particularly critical for qubits with limited decoherence times, as it helps mitigate coherence loss during operations. Therefore, an important problem exists as profiling each qubit pair with its own optimal pulse waveform.

4.2.2 Brute-force Clustering. CR pulses use microwave to trigger Rabi oscillation on the target qubits depending on the state of the control qubit. However, the effect of Rabi oscillation and corresponding errors depend heavily on the physical properties between qubits. For example, a strong drive signal can often lead to coherent errors induced by non-adiabatic dynamics. To capture the interactive Hamiltonian between adjacent qubits, several critical metrics are considered, including the detuning between qubits' frequency, the anharmonicity, and the coupling strength between the qubits. As shown in Figure 6, the transition errors on the control qubit depend strongly on the qubits detuning and the anharmonicity. Apart from the control qubit error, the dominant coefficients in

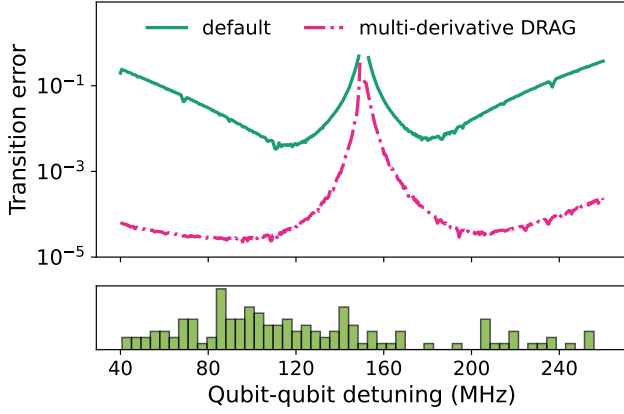


Figure 6: Top: Relationship between remaining transition errors and the frequency detuning for multi-derivative DRAG pulse. The large error in the middle is located at half of the control qubit’s anharmonicity, induced by a two-photon transition. Bottom: Distribution of the qubit-qubit detuning on ibm_brisbane. A few outliers are removed for clarity.

the effective Hamiltonian in Equation 1 are also decided by the frequency detuning and coupling strengths, given by

$$H_{CR} = -\frac{\Delta_{12}}{2}\hat{Z}\hat{I} + \frac{\Omega(t)}{2}(\hat{I}\hat{X} - \frac{J}{2\Delta_{12}}\hat{Z}\hat{X}) \quad (3)$$

where Δ_{12} is the frequency detuning between two qubits and $\Omega(t)$ is the cross resonance drive strength.

The frequency detuning, coupling strength, and anharmonicity of target qubits are prioritized as major features when profiling the optimal CR pulse. Each qubit pair is projected into a multi-dimensional vector, which captures the physical properties that are critical to high-fidelity control. Those vectors are clustered with the Birch algorithm into multiple groups, and partition qubit pairs into sets with similar characteristics. Then one representative qubit pair is selected from each group and three calibration processes are performed on the representatives. The election of the representative allows us to identify the control waveform with optimal fidelity that can be generalized to all pairs within the cluster. To balance calibration efficiency and fidelity, the cluster size is selected to be three, five, and seven. We note that fewer groups result in a faster calibration process, and more groups would lead to a higher accuracy. Taking ECR calibration cost as a reference (1 time unit), the calibration costs of Multi-derivative DRAG and Direct CR are 1.4 and 2.45 time units, respectively. Ideally, the 144 qubit pairs are divided into four subsets as a gap of two edges is required for qubit pairs to be calibrated parallel. And it takes four calibration rounds to calibrate all the connected qubit pairs with a different subset of qubit pairs calibrated in each round. If all qubits are profiled in parallel, the total time taken can be seen as $5 * (1 + 1.4 + 2.46) = 24.3$. In the clustering process, we always select qubit pairs that could be paralleled profiled as representatives. Therefore, the total time can be seen as $1 + 1.4 + 2.46 + 4 * 2.46 = 14.7$, realizing a 1.65 times reduction in the total time. Given that a total of 16 rounds of calibrations are required for real machines, clustering could significantly

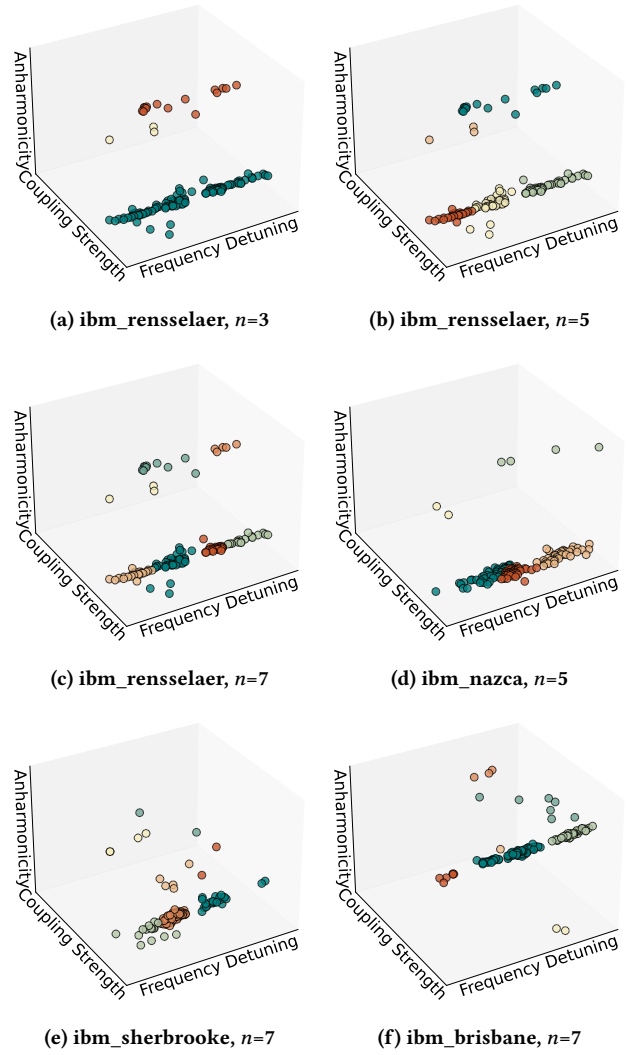


Figure 7: Clustering results for various quantum devices (ibm_renselaer, ibm_nazca, ibm_sherbrooke, and ibm_brisbane) based on frequency detuning, coupling strength, and anharmonicity for various clustering size n .

accelerate the calibration process. We implement this approach on multiple real quantum devices, the results are illustrated in Figure 7. The clustering-based approach optimally distributes calibration resources ensures an effective trade-off between computational efficiency and calibration accuracy across qubit pairs.

4.2.3 Topology-oriented Representative. The Brute-force clustering method manages to strike a balance between calibration cost and the accuracy for waveform profiling. However, the performance of Brute-force clustering depends heavily on the clustering size, which is a hyperparameter predefined, and it is uncertain how many clusters are required to meet the sweet spot of the method for various quantum computer systems.

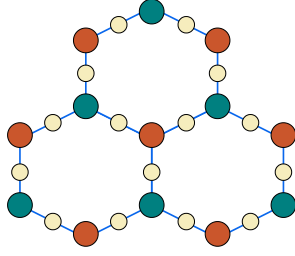
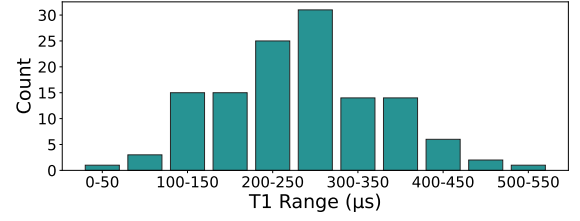


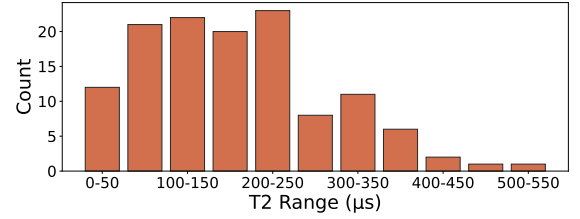
Figure 8: Repetitive pattern of the heavy-hex lattice, colors indicate the pattern of distinct frequencies for control qubits (pale yellow) and two sets of qubits (orange and deep teal).

Recently, the topology of all active IBM quantum devices is based on the heavy-hex lattice, where each unit cell of lattice consists of a hexagonal arrangement of qubits, with an additional qubit on each edge. For CR operations, the resonance frequency executed on the target qubit needs to be precisely off-resonant with neighboring qubit transition frequencies to prevent unwanted interactions, also known as frequency collisions. Compared with the traditional square lattice model, the reduced connectivity of heavy-hex topology brings about fewer frequency constraints, providing a better chance at achieving higher fidelity CR control gates. Previous research has highlighted a recurring pattern among unit cells in the heavy-hex lattice topology [36, 45], where qubits occupying analogous positions across different unit cells exhibit similar physical characteristics. This regularity is a result of the uniform structure of the heavy-hex architecture, where each unit cell maintains a consistent arrangement, leading to shared properties such as frequency detuning, coupling strength, and anharmonicity among qubits in equivalent positions. Figure 8 depicts such a similarity where colors indicate the pattern of distinct frequencies for control and target. Leveraging this structural regularity, each edge in the heavy-hex topology can be classified according to its relative position within the unit cell. By categorizing edges in this way, the calibration complexity is effectively reduced. For each distinct position within the local unit cell, we select a representative edge to undergo the complete three-stage calibration process. This allows us to identify the waveform that achieves optimal fidelity, which can then be generalized to all edges in equivalent positions across the lattice. This approach significantly streamlines the calibration process by minimizing redundant calibrations, making it possible to apply a high-fidelity waveform to all similar edges with greater efficiency. And the cluster size is set to 12 as in a heavy-hex topology composed of hexagons, a hexagon is made up of 12 qubits with 6 on the vertexes and 6 on the sides. Compared with the Brute-force clustering, such a method limits the cluster size to 12 and the classification policy not only includes evident qubit physical parameters but also considers the hidden constraints generated by hardware topology.

4.2.4 Hardware-oriented Policy. Previous solutions have generated waveform profiling for all qubit pairs towards the highest fidelity while limiting a reasonable calibration time. Another critical factor, system knowledge, has been underscored by previous research as essential for optimizing performance on quantum hardware [20],



(a) T1 distribution for ibm_renselaer



(b) T2 distribution for ibm_renselaer

Figure 9: Longitudinal relaxation (T1) and transverse relaxation (T2) distribution for ibm_renselaer, indicating a non-negligible number of qubit pairs with T1/T2 below 150 μs.

though it was not addressed in previous discussions. Typical elements of system knowledge encompass the sequential order of calibrations required to initialize and optimize a qubit, as well as targeted optimizations tailored to specific qubit pairs. The calibration sequence involves channel frequency calibrations and single-qubit calibrations, which are out of the scope of this paper. On the other hand, targeted optimizations could contribute majorly to a more efficient and accurate waveform design. In the hardware-oriented policy, system knowledge of the relationship between frequency detuning and waveform is selected as a major constraint when profiling pulse waveforms. Through numerical simulations, it's exploited that for qubit pairs that reside outside a specific frequency range, calibrating multi-derivative waveform takes much longer time than echoed CR pulse and fails to eliminate error terms to an ideal extent (0.015MHz). Therefore, for such qubit pairs, the echoed CR waveform should be directly adopted to increase the efficiency of the calibration process and improve the calibration accuracy.

Another crucial metric that is missed from previous discussions is the decoherence limit of current quantum computer systems, which exists as an important limit for the duration of quantum programs. The decoherence limit includes the longitudinal relaxation (T1) and the transverse relaxation (T2), which describes how long the excited state and the superposition state of the qubits can be preserved, respectively. As we can observe from Figure 9, for current quantum hardware, the median of T1 and T2 are 269 μs and 172 μs, respectively. A typical pulse of ECR gate takes approximately 665 ns to execute. For an average-quality qubit pair, this implies that the theoretical upper bound on the number of sequential ECR gates that could produce meaningful results is about 258. However, the actual scenario is more severe as the general performance of a quantum program is limited by qubit pairs with the worst performance. For contemporary hardware systems with 127 qubits, the minimal T2 could be only 20 μs, and around 20 qubit pairs bear a

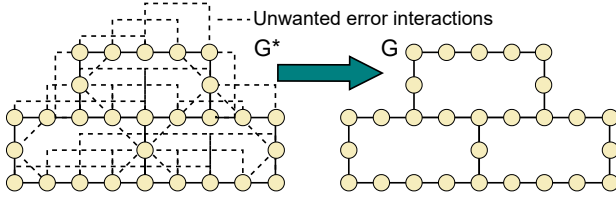


Figure 10: Calibration target: remove unwanted interactions from a connected graph to reach its subgraph, where all edges are calibrated with optimal parameters.

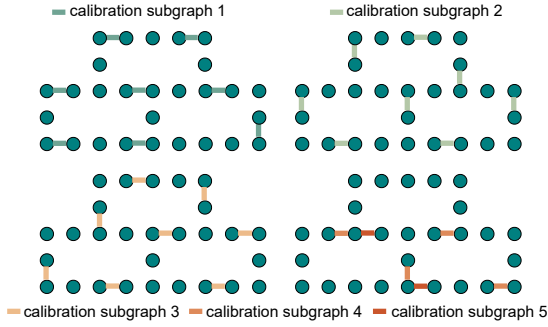


Figure 11: Dividing a heavy-hex coupling graph into five calibration subgraphs for parallel graph traversal.

decoherence limit of less than $60\mu s$. For those qubit pairs, the effectiveness of various waveform strategies is constrained not only by the achievable fidelity but also by the duration of the pulses. Specifically, while an echoed CR pulse may achieve the highest fidelity, its longer duration often renders it less practical than a direct CR pulse. The latter, typically requiring around 60% to 80% the duration of the former, tends to be the preferred choice in scenarios where minimizing pulse duration is critical to overall performance.

In a **hardware-oriented policy**, qubits that fall outside the system limitations and existing system knowledge are profiled using a similar technique to the topology-based approach, as this has proven to be more accurate in identifying problematic qubits. Specifically, qubits with a decoherence time (e.g., T_2) shorter than $85.5\mu s$ (half of the median value) are labeled as **defect qubits**. Meanwhile, qubit pairs with a frequency detuning between 148 MHz and 160 MHz are expected to be more challenging to calibrate with **Multi-derivative DRAG**. By identifying these problematic qubits and qubit pairs ahead of time, hardware resources can be allocated more efficiently, focusing calibration efforts where they are most needed and ultimately improving overall system performance.

4.3 Graph traverse and parallelization

After individually profiling each qubit pair with its optimal CR pulse waveform, a significant challenge remains: calibrating all qubit pairs within a specific time frame without stalling the quantum machine for too much time. A coupling graph with N nodes and E edges is

seen as an undirected graph G . The scheduling problem could be redefined as learning the optimal pulse parameter for some subset of edges in the coupling graph while limiting unwanted interactions outside the subset, which is referred to as $G^* \in G$. Calibration is accomplished by traversing G while calibrating parameters associated with each edge in the target G^* as shown in Figure 10.

As provided in Figure 11, the spacetime structure of the calibration process is defined as calibration subgraphs. Each subgraph represents the part of the graph which is simultaneously calibrated. The separation of calibration subgraphs depends on the elements that could interfere with each during the calibration process. When calibrating CR pulses, it can be seen as executing ECR gates. Therefore, the calibration subgraph could be decided as the maximum number of ECR gates that could execute at the same circuit layer. In general, the total graph G is traversed after the calibration, and at each step, all edges at the subgraph are calibrated. To build the calibration subgraphs, all single paths with a length of one are selected from the coupling map. Then they are separated with the requirement of a minimum distance of two for all edges in a single calibration subgraph. For the heavy-hex structure with 127 qubits, the number of the calibration subgraph could be limited to five, with a maximum number of 38 qubit pairs calibrating simultaneously.

5 Evaluation

5.1 Configuration

All experiments are conducted on Eagle r3 quantum processors provided by IBM quantum service, including `ibm_renselaer`, `ibm_nazca`, and `ibm_strasbourg`. Eagle r3 processor is a family of superconducting-based quantum processing units. They share a configuration of 127 qubits and a heavy-hex coupling graph.

We implement Echoed CR pulses using symbolic functions in Qiskit. In contrast, multi-derivative DRAG pulses and Direct CR pulses are sent to the hardware as arrays of pulse amplitudes. Due to limitations in IBM quantum machines, we split multi-derivative DRAG pulses into two parts to avoid overly complicated custom pulse shapes. We use SciPy [53] to process Hamiltonian tomography results and optimize pulse parameters. Before calibrating ECR gates, we focus on qubits whose single-qubit gate error rates were significantly higher than the device median. These calibrated single-qubit gates are later used in Hamiltonian tomography experiments. We initially set an error threshold of 0.015 MHz for all calibration experiments. If qubit pairs fail to meet this threshold after four calibration rounds, we increase the threshold to 0.3 MHz. Our results show that over 99% of qubit pairs could limit error terms to 0.3 MHz within four calibration rounds. To accurately measure the errors of ECR gates before and after calibration, we employ Interleaved Randomized Benchmarking (IRB) to assess waveform fidelity. Each IRB experiment uses sequence lengths of 1, 10, 20, 50, 100, 150, 250, and 400. We repeat each experiment five times for each pulse configuration to calculate the mean and standard deviation of the gate error. Therefore, the reported gate error represents the average error over the hours following calibration, which has included any drift in system parameters during that time.

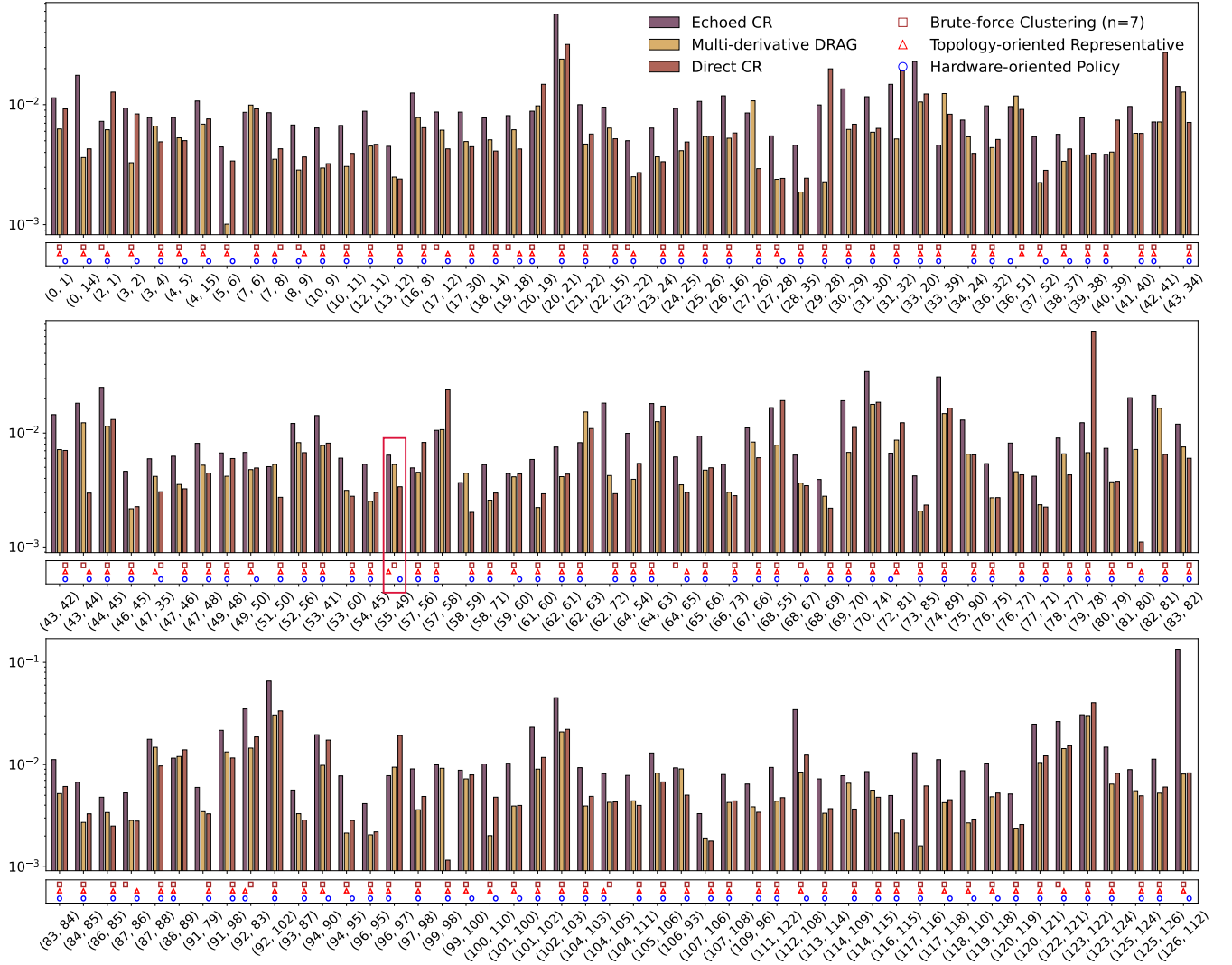


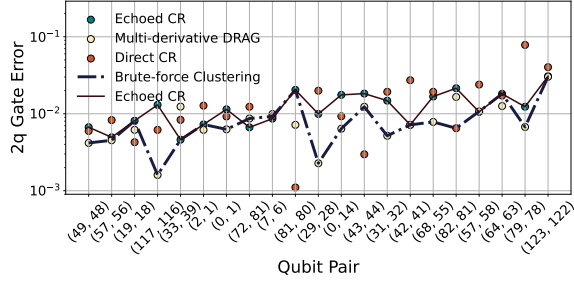
Figure 12: Replaced with a new figure here. Calibration results for `ibm_renselaer` and pulse waveform selections for each qubit pair, which are decided by various policies. For each qubit pair, the fidelities of Echoed CR, Multi-derivative DRAG, and Direct CR are plotted as bars placed from left to right. The waveform selected by each policy is presented by placing geometric patterns in align with the corresponding bars. For example, as marked in the red box, for qubit pair (55, 49), brute-force clustering selects Multi-derivative DRAG, topology-oriented representative selects Echoed CR, and hardware-oriented policy selects Direct CR.

5.2 Gate-level Benchmarking

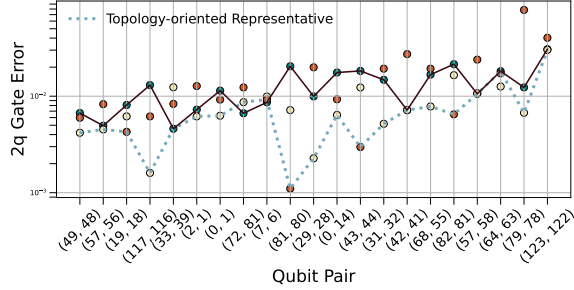
In all policies for profiling optimal ECR gates' waveforms, the primary goal is to achieve maximum fidelity across all qubit pairs. To evaluate whether this can be uniformly accomplished, we performed a comprehensive calibration of three waveforms on each qubit pair, beyond the standard calibration according to each policy.

Benchmarking results, displayed in Figure 12 and Figure 13, include all connected qubit pairs with error terms restricted to 0.015 MHz and 0.03 MHz. The fidelities of Echoed CR, Multi-derivative DRAG, and Direct CR are plotted as bars placed from left to right. The waveform selected by each policy is presented by placing geometric patterns in align with the corresponding bars. For example,

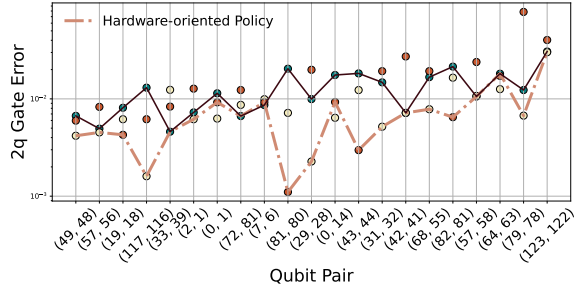
as marked in the red box, for qubit pair (55, 49), brute-force clustering selects Multi-derivative DRAG, topology-oriented representative selects Echoed CR, and hardware-oriented policy selects Direct CR. For Brute-force Clustering ($n=7$), 128 out of 144 qubit pairs (88.9%) achieve optimal fidelity with the profiled pulse waveform. In contrast, the accuracy is elevated to 93.8% under Topology-oriented Representative. When comparing all three profiling policies in Figure 12, the Hardware-oriented Policy achieved nearly the same average fidelity as the Topology-aware Representative (99.40% and 99.39%). For certain qubit pairs, it favored the Direct CR waveform due to its significantly reduced gate duration. For example, the Hardware-oriented Policy selects the Direct CR waveform for qubit



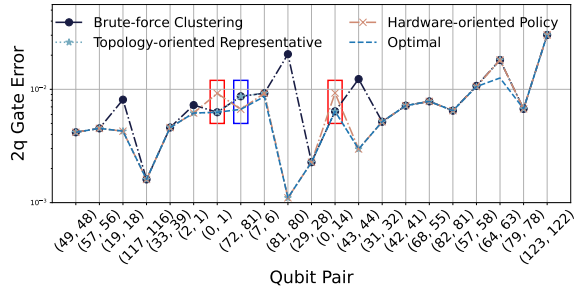
(a) Performance of Brute-force clustering with clustering size of 7. Echoed CR pulse is the default pulse waveform by IBM.



(b) Performance of Topology-oriented Representative.



(c) Performance of Hardware-oriented Policy.



(d) Horizontal comparison of the three profiling policies.

Figure 13: Performance benchmarking of various waveform profiling policies on randomly selected qubit pairs.

pair (0, 1) because qubit 1 has a T_2 time of only $82.99 \mu s$, which is less than half the median value. Direct CR is chosen for the qubit pair (0, 14) since it offers similar fidelity to the Multi-derivative DRAG waveform while maintaining a shorter duration. Meanwhile, for qubit pair (72, 81) with frequency detuning outside the optimal

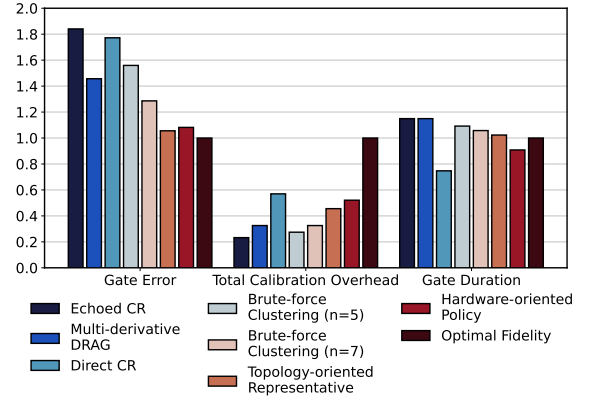


Figure 14: Replaced with a new figure here. The sum of the gate error rate, calibration cost, and gate duration for all qubit pairs based on various profiling policies. All metrics are normalized to the optimal.

range for the Multi-derivative DRAG waveform, the Hardware-oriented Policy automatically selects the echoed CR as the optimal pulse, resulting in a lower error rate. In figure 13, we select 21 representative qubit pairs to further illustrate the results.

To evaluate the performance of different policies, we considered three main metrics for each qubit pair: the sum of error rates, calibration cost, and gate durations. Figure 14 provides a detailed comparison of these metrics, with each normalized to its optimal value. The optimal calibration cost is calculated based on calibrating all three possible waveforms for every qubit pair. The Topology-oriented Representative policy achieves a total error rate close to the optimal scenario, effectively selecting the waveform with the highest fidelity for nearly all qubit pairs. Meanwhile, its total calibration cost is less than half of what it would be if all three waveforms were calibrated for each pair. The Hardware-oriented Policy also attains a similar total error rate compared to the optimal solution. Furthermore, it results in a significantly shorter total gate duration than other profiling policies, which could enable the execution of longer gate sequences within the decoherence limit.

Combining waveform selection for specific qubit pairs and the generalized results for the whole quantum machine, our fine-tuned protocol can achieve almost the optimal error rate on all qubit pairs while limiting the total gate duration. After applying the whole calibration process on the real quantum machine, the medium of the two-qubit gate error rate is reduced to 4.4×10^{-3} , representing a $1.84\times$ improvement compared to the default pulse configuration provided by IBM. For qubits with fabrication defects and shorter decoherence limits, the total pulse duration is reduced by a factor of 1.26. The proposed protocol could reduce the two-qubit gate error rate to the minimum at 1.3×10^{-3} . According to the real IBM quantum hardware results described in the [5], this error rate is already below the two-qubit gate error rate threshold (3×10^{-3}). It means that with our calibration method, the quantum error correction code has entered the region where errors are suppressed.

5.3 Calibration-level Benchmarking

Parallel graph traversal during calibration significantly reduces processing time and enables performance improvements. IBM's current pulse control software has limited support for complex pulse shapes across multiple qubit pairs. While we successfully calibrated 20 qubit pairs using ECR which is IBM's well-supported default option, we encountered difficulties when attempting to calibrate the either of the first four subsets simultaneously. To address this limitation, we divided these subsets into smaller groups of 10 qubit pairs each.

Therefore, calibration subgraphs that contain more than 20 edges are split into smaller graphs with no more than 10 edges each. When splitting the edges, edges that are scheduled for Direct CR calibration are separated from edges scheduled with other calibration methods. So, the long calibration time of Direct CR does not interfere with the runtime of other subgraphs. The calibration subgraphs are then executed sequentially. The Overall Calibration Overhead (T_{total}) can be written as

$$T_{\text{total}} = T_{\text{classical}} + T_{\text{profile}} + N_{\text{groups}} \times T_{\text{CR}},$$

where T_{profile} represents the profiling period and T_{CR} represents the calibration time for each group, which is shorter than or equal to the Direct CR calibration time. Notably, $T_{\text{classical}}$ (the classical algorithm running time) is negligible compared to T_{profile} and T_{CR} . The parallel calibration approach achieved a 7.9 $\times$ reduction in total runtime despite hardware limitations as shown in Figure 15. Without constraints from the devices' software level, the potential speedup could reach 25 $\times$ compared with sequential calibration.

5.4 Device-level Benchmarking

Quantum Volume. Quantum Volume is a single-number metric utilized to characterize the power of quantum computers. The circuit used to measure Quantum Volume is composed of random unitaries involving d qubits with a depth of d . Beginning with a small d , and measured with repetitive trails. Once the measurement outcome satisfies the fidelity requirement, the experiment proceeds with a larger d . The fidelity requirement is shown below

$$\frac{n_h - 2\sqrt{n_h(n_s - \frac{n_h}{n_c})}}{n_c n_s} > \frac{2}{3} \quad (4)$$

where n_h is the number of heavy outputs (outputs with probabilities higher than the median probability), n_c is the number of circuits created which is set to be 100, and n_s is the number of shots. Such a threshold checks if there is at least a 97% chance the heavy output probability is greater than 2/3. Finally, until a square circuit with d depth cannot pass the fidelity requirement, the Quantum Volume is represented as 2^d .

Error Per Layered Gate (EPLG). As current quantum hardware scales to over 127 qubits, Quantum Volume is still a convincing holistic test of processor performance. However, it only contains d qubits with the highest fidelity. The layer fidelity fills the void as the protocol involves as many connected qubits as required. The layer fidelity is measured by splitting selected n qubits into M disjoint layers where each layer contains non-overlapping two-qubit gates and idle qubits. Simultaneous direct randomized benchmarking is conducted on the disjoint layer to measure the error of two-qubit

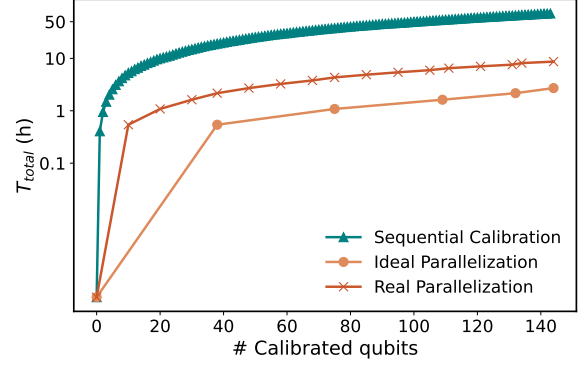


Figure 15: Calibration time comparison among sequential calibration, ideal parallel calibration, and limited parallel calibration due to hardware limitations.

Quantum Device	Quantum Volume		Error Per Layered Gate	
	ECR	Calibrated	ECR	Calibrated
rensselaer	128	256	$(3.08 \pm 0.91) \times 10^{-2}$	$(1.34 \pm 0.29) \times 10^{-2}$
nazca	128	256	$(2.94 \pm 0.83) \times 10^{-2}$	$(1.48 \pm 0.26) \times 10^{-2}$

Table 1: Device-level benchmarking results for ibm_renselaer and ibm_nazca. The EPLG ranges are calculated based on a statistical significance of 95%.

gates and idle qubits. A process fidelity $F_i = \frac{1+(d^2-1)\alpha}{d^2}$ is obtained for each measured decay where $d = 2^n$, and α is the decay rate. The layer fidelity and EPLG is calculated as

$$LF = \prod_{m=1}^M \prod_j F_{j,m} \quad (5)$$

$$EPLG = 1 - LF^{1/n} \quad (6)$$

The device-level benchmarking results for ibm_renselaer and ibm_nazca are shown in Table 1. For both machines, the Quantum Volume is enlarged from 128 to 256, indicating that the quantum processors are able to execute a square circuit with a width of 8 and a depth of 8. Meanwhile, the EPLG for ibm_renselaer is reduced by a factor of 2.3 while EPLG for ibm_nazca also achieves a reduction of 1.99 times, proving the overall improvement in two-qubit gate fidelity. From both local metric (QV) and global metric (EPLG), our optimal pulse profiling technique exhibits great potential in further elevating the performance of current quantum hardware.

5.5 Application-level Benchmarking

To evaluate the real-world effect brought about by the advanced optimal pulse profiling technique, several quantum programs from OpenQASMBench are adopted to examine the application-level performance. Through running programs on noise-free simulators, quantum computer with default pulse setup, and quantum computer with optimal pulse for each qubit pair, the error rate (E) and the

Benchmark	# Qubits	# ECR Gates	ECR Depth	Depth	Error Rate		Fidelity		Error Range	
					Default	Calibrated	Default	Calibrated	Default	Calibrated
adder_n4	4	16	10	41	0.13	0.10	0.87	0.90	±0.004	±0.004
adder_n10	10	146	136	503	0.56	0.49	0.44	0.51	±0.026	±0.017
dnn_n8	8	237	63	249	0.42	0.37	0.71	0.76	±0.019	±0.013
ising_n10	10	90	20	111	0.32	0.24	0.76	0.84	±0.007	±0.008
qpe_n9	9	97	86	339	0.15	0.07	0.94	0.98	±0.012	±0.007
cat_state_n22	22	21	21	66	0.39	0.36	0.61	0.64	±0.013	±0.024
ghz_state_n23	23	22	22	65	0.47	0.41	0.53	0.59	±0.032	±0.019
qram_n20	20	352	225	794	0.74	0.68	0.26	0.32	±0.032	±0.023

Table 2: Application-level benchmarking for ibm_renselaer. Error range is calculated based on a statistical significance of 95%.

fidelity (F) of each circuit are obtained as

$$E = \frac{1}{2} \sum_x |P_{\text{ideal}}(x) - P_{\text{real}}(x)| \quad (7)$$

$$F = \left(\sum_x \sqrt{P_{\text{ideal}}(x) \cdot P_{\text{real}}(x)} \right)^2 \quad (8)$$

where $P_{\text{ideal}}(x)$ represents the ideal probability of outcome x and $P_{\text{real}}(x)$ represents the actual probability of outcome x from real quantum hardware. As demonstrated in Table 2, our optimal pulse profiling technology exhibits error rate reduction and fidelity increase in all benchmarks, with a maximum fidelity increase of 16%, proving its enormous potential in leveraging the hidden power of current quantum hardware.

Our benchmarks spans across wide range of algorithms, qram_n20 is a very complex quantum algorithm we choose to validate our technology on real quantum hardware, and this algorithm has already exceeded the capability of the real quantum hardware and we can observe the default fidelity of qram_n20 is below 30%. For more complicated algorithms, the outcomes are mostly decided upon randomness or decoherence and do not have statistical significance.

Reprofiling Period. For qubits that fall into the specific detuning range and decoherence limit in the hardware-aware policy, the optimal pulse waveforms are constant. For other qubits, we select eight qubit pairs and profile them for four consecutive days, followed by an additional profiling session eight days later. During the first four days, the optimal pulse waveform remains the same. However, eight days later, **five out of eight** qubit pairs experience changes in the optimal pulse waveform. Notably, **four out of five** qubit pairs had at least one single-qubit gate re-calibrated due to drift errors over the eight-day period (maintained by IBM).

6 Related Works

Recent work on optimizing quantum circuits for higher Quantum Volume underscores the importance of gate speed in circuit performance [18]. Techniques to reduce two-qubit gate durations by incorporating pre- and post-single-qubit rotations into the compilation, effectively shortening the gate to its entangling component. Additionally, high-fidelity alternatives to traditional echo pulse sequences, like direct echo-free CX gates leveraging target rotary pulsing [46], have been proposed to eliminate extra single-qubit gates from the duration. These approaches demonstrate that a strategic balance between gate speed and fidelity can enhance overall

circuit performance, however, this approach much more expensive than the ECR gate calibration in terms of the calibration cost.

And previous studies on CR gates have demonstrated significant advancements in mitigating control errors and achieving high-fidelity quantum operations. [27] inspired by the DRAG framework, introduces recursive corrections to mitigate single and multi-photon transition errors, achieving improvement over standard calibration techniques, [9] leverages amplitude control of a microwave drive on one qubit at the resonance frequency of another to generate the desired interaction. However, those method overlooked the limitation on hardware inconsistencies and directly applies the same pulse envelope to all pairs of qubits.

Other SOTA calibration techniques like Floquet [3] is utilized to optimize a single parameter in the control sequence, which can be seen as a part of the calibration process and is already adopted in the calibration of Direct CR (Figure 2a). [26] explores the trade-off between expressivity and calibration overhead in quantum computing instruction set design, instead, we are more care about the individual two qubit gate calibration for each qubits pair, which is orthogonal but more fine-grained. Snake optimizer [20] systematically manages the calibration sequence. For instance, when one qubit pair shows high error, the qubit pair calibration process will sequentially calibrate the qubit frequency, one-qubit gate and then two-qubit gate. Therefore, it is also an orthogonal work and addresses another aspect of the calibration process.

7 Conclusion

In this paper, we present a fine-grained calibration protocol that addresses both hardware variations and parallel execution needs in quantum computers. Our protocol combines three hardware-aware calibration policies with a graph-based method for parallel calibration. Experiments on quantum processors with up to 127 qubits demonstrate significant fidelity improvements while maintaining practical calibration overhead. After optimal pulse profiling, about 20% qubit pairs could achieve an error rate below the QEC threshold [5]. With the heavy-hex topology and qubit number, only a QEC with a distance less than 3 can be realized and probably not utilizing all the high-fidelity qubit pairs. Therefore, real-machine experiments in QEC is largely affected by randomness. However, an effective improvement in QEC could be expected as the improvement in quantum volume and average gate fidelity. This work advances hardware-aware calibration strategies and establishes a foundation for fault-tolerant quantum computing at scale.

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A Artifact Appendix

A.1 Abstract

The artifact contains the source code used to generate, simulate, and evaluate the optimal pulse profiling calibration protocol presented in this paper. Since certain results presented in this work utilized premium quantum hardware that require access tokens and IBM currently suspends its support for pulse-level circuits, this artifact provides a small scale simulation to demonstrate our workflow. A batch script is also provided for users who have access to other quantum hardware platforms with pulse-level access to manually execute them. The artifact provides Jupyter notebooks and python files to reproduce major results in Figure 6, Figure 7, Figure 12, and Figure 14.

A.2 Artifact check-list (meta-information)

- **Program:** Qiskit
- **Run-time environment:** Jupyter Kernel and Python
- **Hardware:** AMD EPYC 9534 64-Core
- **Execution:** Qiskit circuit simulation
- **Output:** Pulse waveform, device information, and fidelity
- **Experiments:** Calibration experiments and randomized benchmarking experiment
- **How much disk space required (approximately)?:** 1 GB to store the artifact directory and python virtual environment.
- **How much time is needed to prepare workflow (approximately)?:** 10 minutes
- **How much time is needed to complete experiments (approximately)?:** 1 hour
- **Publicly available?:** Yes
- **Code licenses (if publicly available)?:** MIT License
- **Workflow automation framework used?:** Jupyter notebook
- **Archived (provide DOI)?:** <https://doi.org/10.5281/zenodo.15104875>

A.3 Description

A.3.1 How to access. The artifact is available on Zenodo <https://doi.org/10.5281/zenodo.15104875>

A.3.2 Hardware dependencies. The artifact relies on quantum circuit simulation available through the Qiskit. Any system which can run python programs should be able to evaluate the artifact.

A.3.3 Software dependencies. The dependencies are listed within requirements.txt.

A.4 Installation

The README.md contains detailed instructions to install the python environment. After downloading the artifact zipfile, and extracting the contents, the environment can be installed via:

```
# cd Adv_Calib_ISCA_Artifact
```

```
# pip install -r requirements.txt
```

The user can then open the jupyter lab with the command:

```
# jupyter lab
```

The jupyter notebook files contain simulation experiments. The python files start with "plot" reproduce results in the paper.

A.5 Evaluation and expected results

The notebook *ecr_calib.ipynb* contains examples how to calibrate and roughly benchmark an echoed cross resonance pulse. The notebook *multi_derivate_calib.ipynb* contains examples how to calibrate and roughly benchmark a multi-derivative DRAG pulse. The notebook *direct_calib.ipynb* contains examples how to calibrate and precisely benchmark a direct CR pulse. The folder *qiskit_utilities* is a custom package tailored for large-scale pulse execution and result analysis on real quantum hardware. The python script *batch_exp.py* can be utilized to parallelly calibrate and benchmark the whole quantum processing unit. The python file *plot_clus.py* reproduces results in Figure 7. The python file *plot_effective_range.py* reproduces results in Figure 6. The python file *plot_t_predict.py* reproduces results in Figure 14. The python file *plot_general_res.py* reproduces results in Figure 12 with real machine data previous obtained from *ibm_rensseleer*.

A.6 Methodology

Submission, reviewing and badging methodology:

- <https://www.acm.org/publications/policies/artifact-review-and-badging-current>
- <https://cTuning.org/ae>